



Fig. 1: IPTV operation process (Courtesy: IPTV India Forum)

infrastructure cost attached with IPTV the most difficult. "Building and providing a complete network, right from developing a strong head-end to providing error-free, last-mile connectivity, is a huge cost for service providers to endure. Also, since IPTV is still in its nascent stage, the return on investment is currently small," says Raju.

Since IPTV uses Internet protocol to send and receive data, at times, television sets can experience pack loss or delays. This can worsen in case of problems related to last-mile connectivity, feels Raju.

Sahu perceives the start-up cost of IPTV as slightly steep, compared to cable TV and DTH, due to the fact that the former requires wires and an IP connection and still needs to be delivered to consumers for as low as Rs 150-250 a month. "This price barrier needs to be crossed," says Sahu, pointing out the brighter side of IPTV for both service providers and consumers. "There are so many interactive services, which can be enabled and used to subsidise or overcome the loss, which one could incur by delivering IP broadcast TV at a low price," he says.

IPTV providers are also concerned that it is heavily reliant on high bandwidth, whereas the broadband access in the country is growing at a slow pace. They are also worried about the fact that the number of IPTV subscri-

ers are still lesser than 50,000, whereas cable TV has 70 million subscribers and DTH 13 million. Due to this, IPTV service gets low priority, in terms of getting content from a lesser number of broadcasters.

The other key issue is the need for a suitable and ubiquitous broadband infrastructure to support the delivery of such services. Just providing stable broadband connectivity to subscribers at high speed, let alone delivering IP-based television content over these connections, is a major challenge for service providers.

The success of IPTV in India depends critically on things such as the unique value proposition it offers in comparison to the existing cable TV fare; successful implementation of last-mile strategies; and successfully ironing out the challenges faced by the new technology. Furthermore, service providers need to emphasise on the quality of services and strengthen customer care.

Technology of the future

"IPTV is a technology of the future and is gradually taking off in India. We firmly believe that IPTV and other value-added services will spur penetration of broadband in India. The IPTV forum has set a target of acquiring 1 million subscribers by 2012. The compelling factors, which will attract

consumers will be IPTV's interactive features and value-added services," states Sahu.

Delivering entertainment content over broadband networks will empower users to view their content at times pre-set by them. IPTV will also serve as a stimulator for the growth of broadband in the country as it will generate the demand for increased broadband capacity and networks.

"Our belief is that IPTV will be more relevant for computer illiterate people in rural and urban India. The level of interactivity provided by IPTV will facilitate better video communication and minimise the need for computer literacy, thereby bridging the digital divide. If the government treats IPTV as an infrastructure service and invests in its setting up, then its growth can reach phenomenal heights. IPTV has the potential to change from a TV medium into an information medium and then into a productivity tool," says Sahu.

Bright potential

Technologies are changing rapidly and so are the innovators of content for these technologies, but what is agonising is that our policy framework is too slow to cope with the changing needs. Customers are more than willing to select any service that entertains them and gives them value for their money. Whether one is withdrawing money from an ATM machine, purchasing a ticket through the Internet, paying utility bills online or buying cinema tickets, user-friendly features are always appreciated and often preferred by consumers. "In such a scenario, the potential for IPTV is quite good, as it can be effectively deployed in multi-development of housing units, malls, government offices, railways, etc," says Prakash.

Of late, receptiveness on the part of the government towards acceptance of new technologies and innovative ideas has been noticed. Less than a year ago, India didn't even have an IPTV policy,